

Guideline/Protocol Title	Pediatric Guidelines for the Diagnosis and Treatment of Tick-Borne Illness
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Committee Review	Antimicrobial Stewardship Subcommittee Pharmacy and Therapeutics Committee
Target Population	Pediatric patients undergoing workup for tick-borne illness
Overview	This document provides evidence-based guidelines for the diagnosis and treatment of tickborne illnesses in pediatric patients
Effective Date	9/2020
Revised Date	10/2022; 10/2024
Expiration Date	10/2027
Schedule for Periodic Review	Every 3 years
Implementation Strategy	See below. Each service will be trained on the guideline.
Education Strategy	ID pharmacist, ID physician, and ASC service line representatives will educate each affected service. Email communication will be sent as well.
Primary Outcome(s)	Selection of antimicrobial therapy and duration of therapy.
Outcome Assessment Plan	
Document Posting Location	☒ CareWeb Guidelines and Protocols ☐ Markey Cancer Center (MCC) Protocols ☒ Other Location: Antimicrobial Stewardship site
Information Technology Needs	☒ Link to LexiDrugs – <i>Medication(s): doxycycline</i>

Pediatric Guidelines for the Management of Tick -Borne Illnesses

Purpose of guidelines: To provide evidence-based information for the treatment of tick-borne infections in pediatric patients admitted to the University of Kentucky Health Care System.

- Goals:**
- To expedite the collection of appropriate diagnostic studies.
 - To assist with early and appropriate diagnosis of tick-borne illnesses.
 - To facilitate early initiation of appropriate antibiotics.

Background:

Annual confirmed cases of Ehrlichiosis and Rocky Mountain Spotted Fever (RMSF) have increased by approximately eight-fold from 2000 to 2018 within the United States (US).^{8,9} Diagnosis of tick-borne illnesses represent a challenge to clinicians as microbiologic testing is often not performed and clinical improvement often occurs prior to laboratory diagnosis. Tick-borne illnesses are a group of important emerging diseases that can have various clinical presentations. Additionally, ticks have been known to harbor multiple pathogens which can lead to simultaneous infections. Kentucky has a significantly higher incidence of RMSF and ehrlichiosis when compared to other common tick-borne illnesses (*Babesia*, *Borrelia*, *Anaplasma*, etc). Both RMSF and ehrlichiosis are well known to cause severe morbidity, including but not limited to meningoencephalitis, cardiovascular failure, hepatic injury, acute respiratory distress syndrome, and renal failure.⁷

Diagnosis and Initial Work up of Tick-borne illnesses

Tick-borne illnesses can be difficult to diagnose and often present with non-specific symptoms. The classic triad of symptoms associated with tick-borne illness includes fever, headache, and rash. The most common presentation includes an initial prodrome of flu-like illness including fever, myalgia, headache, malaise, nausea, vomiting, and diarrhea. Although the majority of tick-borne illnesses occur between April and October, transmission can occur year-round in warmer climates. It is important to obtain complete exposure history including: (1) history of tick bite, (2) exposure to environments with ticks, (3) identification of clusters of illness among contacts, and (4) travel history to endemic areas. Although history of tick exposure is important for diagnosis, lack of tick bite cannot serve as a diagnostic rule out. It is important to ascertain onset of illness and presence of rash. Although testing is important, the decision to initiate treatment is commonly based on a clinical presumptive diagnosis made through clinical findings and epidemiological risk factors. For patients undergoing a work-up for tick-borne illness, treatment should not be withheld pending results of diagnostic tests (i.e. therapy should be initiated promptly and should **not** be discontinued until appropriate diagnostics return negative).

Infectious Diseases (ID) Consultation:

An ID consultation is **recommended** in the following patients and/or clinical scenarios:

- Pregnant patients
- Pediatric patients
- Immunocompromised patients
- Patients with a life-threatening allergy to doxycycline
- Lack of defervescence or improvement after >48 hours on doxycycline
- Suspicion for Babesiosis
- Suspicion for disseminated (neurological, cardiac, arthritic) Lyme disease
- Equivocal diagnostic testing

Pediatric Concerns:

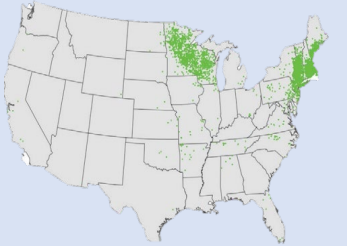
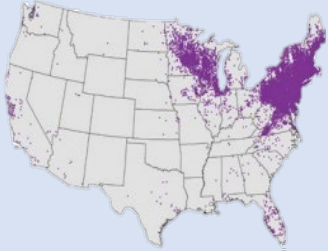
Although tetracycline has historically been associated with teeth discoloration, doxycycline has not demonstrated the same affinity for calcium binding. Recent comparative data suggest that doxycycline is **not** likely to cause visible teeth staining or enamel hypoplasia in young children (<8 years of age). The revised recommendation by the American Academy of Pediatrics, reflected within the 2018 *Red Book*, is that **doxycycline may be administered for durations ≤21 days without regard to the patient's age.**³ Doxycycline is the drug of choice for treatment of human ehrlichiosis and anaplasmosis and should be used **regardless** of patient age.³

Figure 1: Acute Tick-Borne Illness Testing and Treatment Algorithm



(a) See Table 4 for SCM order set information; (b) Place in sterile, airtight container; (c) Consider obtaining *Anaplasma* antibodies if exposure occurred >1 week prior (d) Pediatric dosing: Doxycycline 2.2mg/kg/dose IV/PO q12h (maximum dose: 100 mg); (e) Patients receiving doxycycline should be counseled to avoid extended sun exposure due to risk of phototoxicity and take doses with food to decrease associated GI toxicity.

Table 1: Clinical presentation of Selected Tick-Borne Illnesses^{1,2}

Disease	Vector	Endemic Region(s)	Incubation	Laboratory Indices	Initial Presentation	Cutaneous Features
Rocky Mountain Spotted Fever	American Dog tick (<i>D. variabilis</i>) Wood tick (<i>D. andersoni</i>)		3-12 days	– Normal or ↑ WBC – ↑ ANC – ↑ Transaminases – Thrombocytopenia – Hyponatremia	Fever, chills, severe headache, myalgia, nausea, vomiting, diarrhea, anorexia	Maculopapular rash approximately 2-4 days after fever onset in most; rash may become petechial and involve palms and soles
Human Monocytic Ehrlichiosis	Lone Star tick (<i>A. americanum</i>) Blacklegged tick (<i>I. scapularis</i>)		5-14 days	– ↓ WBC – ↑ Transaminases – Hyponatremia – Thrombocytopenia	Fever, headache, myalgia, nausea, diarrhea, vomiting, confusion, rash	Variable pattern that appears ~5 days after illness onset; may involve palms and soles <i>Rash present in ~30% of adults and ~60% of children</i>
Anaplasmosis	Blacklegged tick (<i>I. scapularis</i>)		5-14 days	– ↓ WBC – ↑ Transaminases – Thrombocytopenia	Fever, headache, myalgia, nausea, diarrhea, and vomiting	Rare (<10%)
Babesiosis	Blacklegged tick (<i>I. scapularis</i>)		≥1-9 weeks	– ↑ Transaminases – ↓ Hematocrit – ↓ Reticulocytes – Thrombocytopenia	Chills, headache, myalgia, arthralgia	None
Lyme Disease Early Stage	Blacklegged tick (<i>I. scapularis</i>)		3-30 days	None	Fever, myalgia, headache, nausea, fatigue	Erythema migrans
Lyme Disease Disseminated			≥2 weeks	None	Cardiac or musculoskeletal manifestations: See Table 2	None
			≥2 weeks	None	Neurologic or musculoskeletal manifestations: See Table 2	None

Images obtained from: <https://www.cdc.gov/ticks/tickbornediseases/overview.html>

For more information about tick-borne illnesses, please visit the CDC website: <https://www.cdc.gov/ticks/tickbornediseases/index.html>

Lyme Disease:

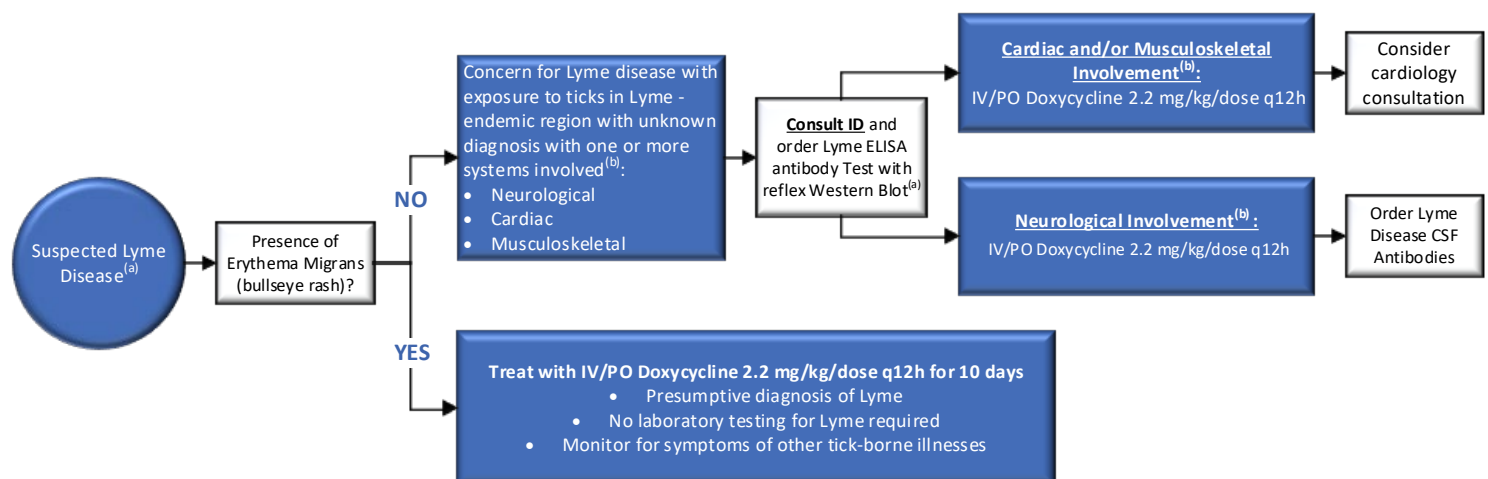
Per the CDC, **Kentucky is defined as a low Lyme incidence state** with <10 confirmed cases/100,000 over the last three reporting years.^{17,18} Per the Kentucky Department of Public Health, there were only six cases of confirmed Lyme Disease in Kentucky in 2017, with the majority of these cases located in southern and western Kentucky.¹⁹

- It is crucial that an accurate history be ascertained in patients with suspicion for Lyme disease. The over-diagnosis of disseminated Lyme disease can lead to inappropriate use of health services and avoidable treatment-related illness.¹⁶
- For patients who have neurological symptoms, the American Academy of Neurology and the Infectious Disease Society of America recommend parenteral regimens for the treatment of neuroborreliosis.²⁰
- There have been European studies that suggest oral doxycycline and parenteral ceftriaxone may be equally effective in the treatment of neuroborreliosis with facial palsy with no significant differences in minimal inhibitory concentrations (MIC) among different species; however, this remains untested in the US.²¹
- Given the complexities associated with the diagnosis of disseminated Lyme disease, **ID consultation is advised.**

Table 2: CDC Surveillance Definitions of Lyme Disease¹⁷

Organ System	Definition	Caveats to Definition
Skin	Erythema Migrans (EM): Expanding skin lesion, often with partial central clearing (bullseye appearance) around tick bite. Primary Lesion must be ≥ 5cm across largest diameter and diagnosis of EM must be made by physician.	Annular erythematous lesions occurring within hours of tick bite represent hypersensitivity reactions and do NOT qualify as EM.
Musculoskeletal System	Recurrent, brief attacks (weeks or months) of objective joint swelling, sometimes followed by chronic arthritis	Chronic progressive arthritis not preceded by brief attacks, chronic symmetrical polyarthritis or arthralgia, myalgia, or fibromyalgia syndromes alone are NOT criteria for musculoskeletal involvement.
Central Nervous System (CNS)	Any of the following, alone or in combination: - Lymphocytic meningitis - Cranial neuritis, particularly facial palsy (may be bilateral) - Radiculoneuropathy - Encephalomyelitis	- Headache, fatigue, paresthesia, or mildly stiff neck alone, is NOT criteria for neurologic involvement. - Encephalomyelitis must be confirmed by demonstration of antibody production against <i>Borrelia burgdorferi</i> in the cerebrospinal fluid (CSF), evidenced by a higher titer of antibody in CSF than in serum.
Cardiovascular System	Acute onset of high-grade (2 nd or 3 rd degree) atrioventricular (AV) conduction defect that resolves in days to weeks.	Palpitations, bradycardia, bundle branch block, or myocarditis alone is NOT criteria for cardiovascular involvement.

Figure 2: Diagnosis and Treatment of Suspected Lyme Disease



(a): Suspicion should only be in those with recent travel to a Lyme-endemic area and symptomatology consistent with Lyme disease (see Table 2); (b): See Table 2 for more information on diagnostic criteria for cardiac, neurologic, and musculoskeletal involvement; (c): Pediatric dosing: IV/PO Doxycycline 2.2mg/kg/dose q12h (max dose: 100 mg); Patients receiving doxycycline should be counseled to avoid extended sun exposure due to risk of phototoxicity and take doses with food to decrease associated GI toxicity.

Table 3: Selected Laboratory and treatment regimens for Tick-Borne Illnesses^{1, 5, 12, 20}

Disease	Diagnostic Test(s)	Laboratory Confirmation And Timing Interpretation	Treatment	Other Considerations
Rocky Mountain Spotted Fever^a <i>Rickettsia rickettsii</i>	Acute and convalescent serology	1. 4-fold increase in antibody titer • IgM present day 3-8 and peaks at 1 month (present 3-4 months) • IgG present within 3 weeks and peaks at 1-3 months (present >12 months)	Treatment^{c,d}: Doxycycline 2.2 mg/kg/dose IV/PO Q12h Max dose: 100 mg Duration: 5 days; <i>Treat at least 3 days after clinical improvement and fever resolution</i>	- Serology is preferred test of diagnosis as <i>Rickettsia</i> targets endothelial cells. - PCR may be positive in advanced disease - RMSF cannot be confirmed using single antibody test
Human Monocytic Ehrlichiosis^a <i>Ehrlichia chaffeensis</i>	<i>E. chaffeensis</i> PCR from blood	1. <i>Chaffeensis</i> DNA detected 2. 4-fold increase in antibody titer* 3. Morulae detected and antibody titer positive*	Treatment^{c,d}: Doxycycline 2.2 mg/kg/dose IV/PO Q12h Max dose: 100 mg Duration: 5 days; <i>Treat at least 3 days after clinical improvement and fever resolution</i>	- CSF findings typically include pleocytosis and protein elevation - Negative PCR does <u>not</u> rule out disease
Human Granulocytic Anaplasmosis^a <i>Anaplasmosis phagocytophilum</i>	<i>A. phagocytophilum</i> PCR from blood	1. <i>A. phagocytophilum</i> DNA detected 2. 4-fold increase in antibody titer* 3. Morulae detected and positive antibody titer	Treatment^{c,d}: Doxycycline 2.2 mg/kg/dose IV/PO Q12h Max dose: 100 mg Duration: 10 days ^(e)	- CSF is usually normal - Negative PCR does <u>not</u> rule out disease - Diagnosis cannot be confirmed with single antibody test
Babesiosis <i>Babesia microti</i>	Peripheral blood smear of intraerythrocytic babesia parasites (stat) <i>Babesia</i> PCR <i>Babesia</i> antibody titer by IFA (supportive)	1. Identification of <i>Babesia</i> organisms in RBC 2. <i>Babesia</i> DNA detected • Antibodies present within 4 weeks of onset of parasitemia	Treatment: Consult ID Duration: 7 days	ID consult strongly recommended - Treatment recommended in symptomatic patients with PCR positive or microscopic identification
Lyme Disease^b <i>Borrelia burgdorferi</i>	ELISA with Western Blot	1. Acute and/or convalescent antibody IgG serology in serum 2. <i>B. burgdorferi</i> DNA in CSF or synovial fluid • Antibodies develop 2-4 weeks after tick bite • Acute and/or convalescent antibody IgG antibodies appear >4 weeks following tick bite	<u>Erythema Migrans</u> Treatment^{c,d}: Doxycycline 2.2 mg/kg/dose IV/PO Q12h Max dose: 100 mg Duration: 10 days	ID consult strongly recommended - Patients with Erythema Migrans (bullseye rash) at presentation should be considered to have presumptive diagnosis of Lyme Disease. - CSF and synovial fluid testing are <u>not</u> diagnostic without concurrent serum testing - Consider cardiology consultation for cardiac involvement - See IDSA Guidelines for Diagnosis and Treatment of Lyme Disease for additional information regarding diagnostics
			<u>Lyme Arthritis</u> Treatment^{c,d}: Doxycycline 2.2 mg/kg/dose IV/PO Q12h Max dose: 100 mg Duration: 28 days	
			<u>Lyme Carditis</u> Treatment^{c,d}: Doxycycline 2.2 mg/kg/dose IV/PO Q12h Max dose: 100 mg Duration: 14 days	
			<u>Neurological Lyme</u> Treatment^{c,d}: Doxycycline 2.2 mg/kg/dose IV/PO q12h Max dose: 100 mg Duration: 14 days	

(a): Delay in treatment may result in severe illness or death; (b): An ID consult is strongly recommended if Babesiosis or cardiac, neurologic, or musculoskeletal Lyme disease is suspected; (c): The American Academy of Pediatrics recommend oral doxycycline, when possible; *Antibodies may not be present in the first week of disease; (d): Patients receiving doxycycline should be counseled to and avoid extended sun exposure due to risk of phototoxicity and take doses with food to decrease incidence of nausea/vomiting; (e): Treatment duration for anaplasmosis is prolonged at 10-14 days to provide appropriate therapy for possible concurrent infection with *Borrelia spp.*

Table 4: Diagnostic Order Identification in Epic:

Diagnostic	Epic order
Identification of Tick Species	"Arthropod Identification"
Rocky Mountain Spotted Fever Acute and Convalescent Serology	"Rickettsia (RMSF) Ab IgG and IgM w/ Reflex Titer"
<i>Anaplasma phagocytophilum</i> and <i>Ehrlichia chaffeensis</i> PCR	"EHRLICHIA AND ANAPLASMA SPECIES BY PCR (SO)"
Serum <i>Ehrlichia</i> Antibodies	"EHRL Chaffeensis AB"
<i>Anaplasma</i> and <i>Ehrlichia</i> Peripheral blood smear	"Peripheral Blood Smear" <i>Click "Add Comments" → Specify Anaplasma and Ehrlichia</i>
Lyme ELISA with Western Blot	"Borrelia burgdorferi VlsE1/pepC10 Antibodies, total by ELISA with reflex to IgG and IgM by Elisa (modified Two-Tier Testing) SO"
Lyme Disease CSF antibodies	"Borrelia species by PCR (Lyme Disease) (SO)"

Tick Bite Prevention:

Before going outside in a tick endemic area, it is prudent to know where ticks are found. Ticks tend to live in grassy, wooded, or bushy areas, with that said one should be mindful that many tick bites happen in peoples own yards.¹⁰ Ticks can be carried into the house on pets, clothing, gear. If clothing does not require washing, it should be tumbled in a dryer set on high for ten or more minutes to kill ticks. If clothes require washing first, hot water is recommended.¹⁰ Following potential exposure, individuals should check the entire body for ticks and shower as soon as possible to reduce the chances of becoming ill. When performing body checks, it is pertinent to understand that ticks prefer skin folds or areas covered in hair (examples: in or around the ears, in the hair, between the thighs, under the armpit, etc).

Tick Bite Prophylaxis:

Before going outdoors, the CDC recommends treating clothing and gear with products containing 0.5% Permethrin. Other precautions include insect repellents such as products containing DEET, picaridin, IR3535, Oil of Lemon Eucalyptus (OLE), para-menthane-diol (PMD), or 2-undecanone.¹⁰ While outdoors, individuals should attempt to avoid excessive contact with environments known to have a high tick burden such as tall grasses or brushy areas, additionally sticking to hiking trails is also helpful. Regarding pets, vaccines for ticks are not available for most tickborne diseases. The CDC recommends talking to your veterinarian regarding the best prophylactic products for dogs and generally recommends against applying tick prevention products to cats as they are extremely sensitive to many chemicals.¹¹ It is also a good idea to reduce tick habitat in your yard, and after an exposure, check your dogs (examples of common areas: behind the ears, in-between the toes, and in the groin). For prevention of Lyme disease after recognized tick bite, routine use of prophylaxis with antimicrobials or serologic testing is not recommended at this time.¹ In areas with high endemic rates of Lyme disease, a single dose of 200mg Doxycycline can be considered for high risk bites with tick that has been attached for ≥36 hours.

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